MSc Management Consultancy · UCD Michael Smurfit Business School

AI-Driven Predictive Maintenance in the Utilities Sector

Benchmarking and Opportunity Assessment
Irish and European Electricity, Water and Oil & Gas Markets

April 2026

EXECUTIVE SUMMARY

From Industry Strain to AI-Enabled Resolution

THE LANDSCAPE

€37M CML penalty exposure ESB Networks 2021–23	1,266 GWh wind curtailed EirGrid 2024 (14%)	€20M CRU leakage penalty Uisce Éireann 2024	95,000 customers under Boil Water Notices

THE OPPORTUNITY

ESB Networks

€37M CML exposure addressable

- **Metric:** 64,754 Power Outages.
- **Driver:** LSTM/RF monitoring to cut CI/CML penalties.
- **Policy:** CRU PR6 compliant; AI as non-wire Opex.

EirGrid

€6.6M TSO incentive uncaptured

- **Trend:** 1,266 GWh Wind Curtailed (↑ YoY).
- **Driver:** 93.7% LSTM Forecasting Accuracy.
- **Value:** Closes DS3 Balancing Cost Gap.

Uisce Éireann

€20M leakage penalty + 40% NRW

- **Metrics:** 36% Leakage | 95k under BWN.
- **Driver:** RF Pump Fault.
- **Impact:** Meets NRW targets & treatment compliance.

THE BUSINESS CASE

MODEL 1 · LSTM · Time-Series Grid Assets

Target: HV/EHV Fleet (ESB Networks & EirGrid)
Rationale: LSTM captures temporal degradation (DGA/thermal) where RF fails.
Impact: 93.7% Accuracy (+12.5% vs ARIMA), 48% failure reduction, €21M 5-year NPV (€3.3M–€8.3M annual savings)

&

MODEL 2 · RANDOM FOREST · Rotating Equipment

Target: Rotating Assets (ESB Networks & Uisce Éireann)
Rationale: RF classifies discrete fault states (Normal/Warning/Critical); SHAP values ensure EU AI Act explainability.
Impact: 97.2% Detection Accuracy and 15% Downtime Reduction(€500K avoided across 7 Sitech models)

COMBINED BENEFIT · LSTM + RF together address all three utility penalty exposures · Total 5-year NPV: up to €35M · Payback: 14–32 months · CRU-compliant as non-wire Opex

accenture

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Section 01

The Landscape

Of AI Predictive Maintenance in The Utility Sector

- Utilities face rising financial and regulatory pressure across assets and operations
- AI-driven predictive maintenance is proven, with strong global performance benchmarks
- Sector maturity varies, with electricity scaling and water lagging but high-potential
- The barrier is not technology, but data readiness and organisational execution

O&G leads. Electricity is scaling. Water is emerging.

The commercial case is established, the question is where Ireland sits.

EARLY — WATER — ELECTRICITY — OIL & GAS — ADVANCED

WATER · EMERGING

Industry projections

- Industry benchmarks suggest 10–40% maintenance cost reduction potential
- Downtime reduction varies by asset type and data maturity
- Primary constraint: inconsistent sensor coverage and data quality

PdM — Predictive Maintenance

Using AI and sensor data to anticipate equipment failure before it occurs.

ELECTRICITY · SCALING

Vistra Corp (USA) — published case

- 400+ AI models deployed across generation fleet
- 340,000 tonnes CO_2 abated per year
- Substation deployments report forced outage reductions of 20–25% (industry case studies)

Vistra Corp Sustainability Report (2023); EirGrid GAR

OIL & GAS · MOST MATURE

Shell · BP · ADNOC · Chevron

- Shell and BP report 20% downtime reduction from AI-enabled PdM
- ADNOC cites $500M operational value
- ROI of 2–6 months achievable under favourable conditions

(McKinsey, Deloitte, BCG analyses)

McKinsey (2021); BCG (2025); Zapium (2026)

McKinsey (2021); IWA (2022); Vistra Corp (2023); BCG (2025); Zapium (2026)

Five AI architectures — each solving a distinct problem across Irish electricity, water and gas assets

1	**LSTM**	HV Transformers — ESB Networks / EirGrid	· Research indicates 93.7% accuracy vs 81.2% for ARIMA baselines · Significant reduction in physical inspection frequency · Lifespan extension of 5–7 years cited in literature (asset-specific)	*Mirshekali et al. (2023), Energies; Nunes (2025)*
2	**Autoencoders VAE**	Smart Meters / BESS — ESB / EirGrid / Uisce Éireann	· 4–21% improvement in non-technical loss detection · 4–13% false alarm reduction · Electricity theft estimated at $6B annually in the US market alone	*Takiddin et al. (2022), IEEE Systems Journal 16(3)*
3	**Random Forest**	Generation Plant — ESB / EirGrid / Uisce Éireann pumps	· Fault classification accuracy of 85–92% reported across generation plant datasets · Siemens Omnivise deployment at Schluchseewerk AG provides real-world validation	*Dritsas & Trigka (2025)*
4	**XGBoost**	Grid Dispatch — EirGrid / ESB Networks	· MAPE of 3.56% on electricity load forecasting, outperforming ARIMA baselines · Hybrid XGBoost-LSTM models show further accuracy gains over either model alone	*Kanwal (2024); Romanuke (2025)*
5	**Transformer**	National Grid — EirGrid demand planning	· Lower MAE and MSE than LSTM and ARIMA on standard benchmarks (ETTh1, ETTm1, Weather) · Particularly relevant for high-renewable grids	*Wu et al. (2021), NeurIPS*

LSTM
Recurrent neural network learning temporal patterns in sensor time-series data.

MAPE — Mean Absolute % Error
Forecasting accuracy measure, lower values indicate predictions closer to actual.

MAE / MSE
Statistical error measures for prediction model accuracy — lower values are better.

Mirshekali (2023); Takiddin et al. (2022); Dritsas & Trigka (2025); Kanwal (2024); Wu et al. (2021)

The financial case is not theoretical.

Industry case studies demonstrate measurable returns across multiple sectors.

POWER GENERATION · USA

AI thermal monitoring — generation fleet deployment

Reported savings per line / yr	**$500K+**
Facility-level savings at scale	**$1M+**
Cost of unplanned outage	**$125K/hr**

400+ AI models deployed across fleet

AUTOMOTIVE MANUFACTURING

Deep learning vibration analysis for bearing fault detection

Downtime reduction (reported)	**50–70%**
MTBF improvement	**Significant**
Bearing detection accuracy	**>95%**

18-month payback achievable in favourable conditions

Illustrative — manufacturing sector benchmarks (consulting analyses)

CHEMICAL / INDUSTRIAL

Asset Health Centre — plug-and-play model library

Savings across multiple models	**€500K**
Single pump pilot saving	**€60K**
Downtime cost avoided	**€500K/day**

22 factories — scalable model library

Sitech deployment — PwC / Mainnovation PdM 4.0 Framework

MTBF — Mean Time Between Failures
Average operating time between equipment failures, higher indicates greater reliability.

PwC / Mainnovation PdM 4.0 Framework; Vistra Corp (2023); industry case study literature

In Ireland, the driver is not cost savings alone.

It is compliance, resilience and regulatory obligation.

40–43%

of water produced is lost to non-revenue water among the highest rates in the EU. Uisce Éireann and EPA Ireland report this consistently across annual assessments.

Uisce Éireann Annual Report; EPA Ireland

4–13%

potential energy savings available through AI-optimised pump scheduling and process control in water treatment and distribution (IWA, 2022).

International Water Association (2022)

NIS2 · EPA

NIS2 Directive, applicable from October 2024, requires robust cybersecurity for operational technology in essential services.

NIS2 Directive (EU) 2022/2555; EPA Ireland

NRW — Non-Revenue Water
Water produced but lost before billing, through leaks, theft, or metering errors.

SCADA
Industrial control systems managing water treatment and distribution infrastructure.

Priority assets — where PdM is immediately viable

HIGHEST SCADA COVERAGE

Pumping Stations

High failure cost per event - strong data availability

ENERGY CRITICAL

WWTP Aeration Blowers

30–40% of WWTP operational expenditure

HIGHEST REGULATORY RISK

Chlorination Systems

Critical public health asset - strict EPA compliance threshold

GEOSPATIAL MODEL

NRW Leakage Detection

40–43% of water lost — ML-based geospatial detection viable

Uisce Éireann Annual Report; EPA Ireland; IWA (2022); NIS2 Directive (EU) 2022/2555

The barriers are known.

Organisations that address them in sequence are those that scale.

Source: Mainnovation PdM 4.0 Industry Survey — % of maintenance and operations professionals citing each factor as a significant barrier to PdM adoption across European industrial sectors.

Data readiness — 67%

Inconsistent sensor coverage and legacy records limit immediate AI applicability

Budget & business case — 60%

Robust ROI cases required before organisations commit capital beyond pilot stage

Data security & cybersecurity — 59%

NIS2 obligations from October 2024 add regulatory urgency to this barrier

Technology readiness — 47%

Legacy infrastructure gaps require investment before AI models can be deployed

Culture & trust — 35%

Executive sponsorship and workforce training separate scalers from pilot-mode organisations

The sequencing rule

1. Fix data infrastructure first
2. Build the ROI case with a pilot
3. Harden cybersecurity in parallel
4. Scale with executive sponsorship

PdM Maturity
Framework measuring AI maintenance capability, from manual inspection (L1) to cognitive AI (L4).

Key insight
The algorithm is rarely the problem. Data readiness and organisational commitment determine whether industry benchmark outcomes are achievable in practice.

Section 02

The Opportunity

for AI Predictive Maintenance in The Utility Sector

- €60M+ exposure across Irish utilities creates a clear and immediate business case
- AI models align directly to asset-level failures and operational inefficiencies
- Proven ROI through penalty avoidance, cost reduction, and performance optimisation
- Deployment is feasible now within existing regulatory and Opex frameworks

Three utilities. Three distinct regulatory exposures.

The penalty exposure alone justifies the business case for AI-driven predictive maintenance.

ESB NETWORKS

€37M

CML penalty exposure
2021–23

€389.7M distribution Opex
€7.3B distribution asset base

EIRGRID

€6.6M

TSO incentive revenue
left uncaptured

€1,050M allowed revenue (2023)
1,266 GWh wind curtailed

UISCE ÉIREANN

€20M

CRU leakage penalty
2024

€1,606M revenue (2024)
38% national leakage rate

Why this matters for AI Each exposure is reducible via AI-driven PdM; CRU permits AI as Opex.

CRU Revenue Reports; ESB Networks Annual Report (2023); EirGrid GAR (2023); Uisce Éireann Annual Report (2024)

Transformer Health & Asset Protection

Long Short-Term Memory networks identify degradation patterns months before catastrophic failure.

INDUSTRY BENCHMARK · UK TSO

Failure rate	Baseline	→	**–48%**	*reduction*
Annual Opex	£3.8M	→	**£2.5M**	*–35%*
ROI payback period	—	→	**11 wks**	*From deploy*
Forecast accuracy	Manual	→	**92%**	*90-day horizon*

89%
of transformer failures had detectable precursor signals that manual inspection missed

€12.3M *ESB, CRU 2021–23*
avg annual CML penalty

2.5–4 yrs *Eurelectric 2025*
transformer lead time

£1.2M *UK grid benchmark*
per transformer replaced

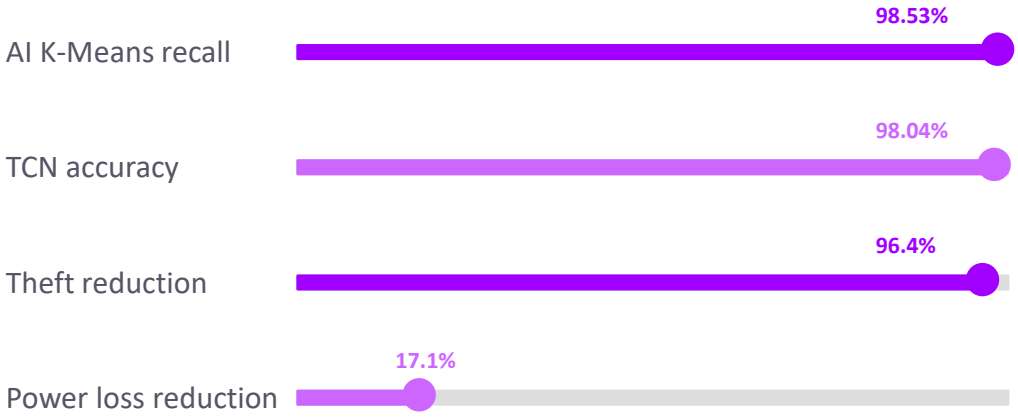
Qualitative impact Shorter outages protect ESB's 814,816 vulnerable customers.

UK TSO anonymised case study; Iberdrola Corporate Report; Eurelectric (2025); CRU CML Decision; Mirshekali et al. (2023)

Cybersecurity & Non-Technical Loss Detection

Anomaly detection across 1.9M smart meters — serving dual regulatory and financial imperatives.

AMI ANOMALY DETECTION — BENCHMARK PERFORMANCE



$96B
global non-technical loss annually
· Khan et al. (2025)

No published Irish-specific NTL figure €20M CRU leakage penalty to Uisce Éireann is the closest domestic signal.

NIS2
MANDATORY
October 2024 · In force now

NIS2 mandates robust OT cybersecurity; autoencoders fulfil this.

1.9M *End-2024, full AMI*
ESB smart meters deployed

22% *Consumed by data centres*
of Irish metered electricity

Khan et al. (2025); Takiddin et al. (2022) IEEE Systems Journal; NIS2 Directive 2022/2555; ESB Networks AMI Report (2024)

Fault Classification & Dispatch Forecasting

One model diagnoses what will fail. The other optimises how the grid responds.

RANDOM FOREST

Fault Classification

97.2% — Detection accuracy — *GM bearing dataset*

15% — Downtime reduction — *Manufacturing benchmarks*

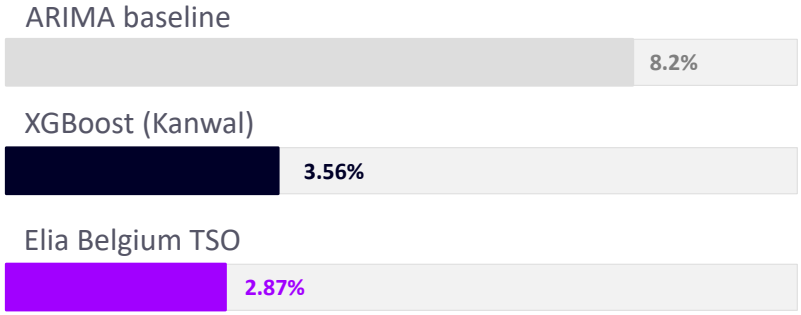
€500K — Cost avoided / 7 models — *Sitech — 22 factories*

GM = automotive; Sitech = chemical industry — not utility-sector specific.

XGBOOST

Dispatch Forecasting

FORECAST ERROR (MAPE) — LOWER IS BETTER



Irish anchor · €47M DS3 payments (Dec 2023). €6.6M TSO incentive uncaptured. Better forecasting closes the gap.

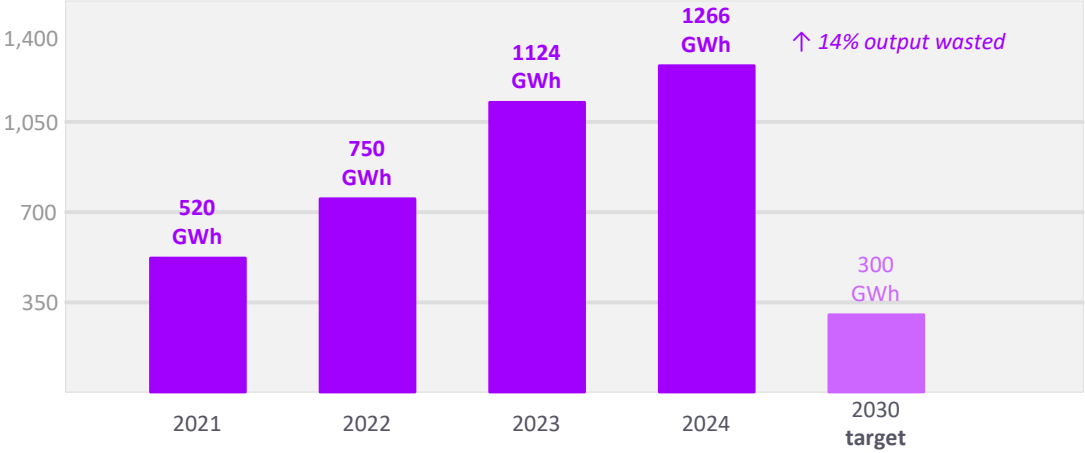
CRU Agile Investment Framework — Permits AI-enabled PdM as non-wire Opex — no new capex required.

Dritsas & Trigka (2025); Sitech / PwC Mainnovation PdM 4.0; Kanwal (2024); Elia Belgium; CRU Agile Investment Framework

Long-Horizon Forecasting & Renewable Curtailment Reduction

Ireland's wind curtailment rate is rising. Autoformer provides the forecasting precision to reverse it.

IRELAND — WIND CURTAILMENT TREND



1% curtailment reduction = 137 GWh recovered · Autoformer enables proactive dispatch optimisation to approach the 2030 target

2030 target · SNSP 95% by 2030

Each 1% curtailment cut = 137 GWh recovered

MODEL PERFORMANCE

MAE / MSE accuracy

Outperforms LSTM + ARIMA on all standard benchmarks

Wu et al. (2021) — ETTh1, ETTm1, Weather

Contingency screening

Thousands of scenarios per minute

RTE France — continental grid planning

Wu (2021) uses Chinese ETT; RTE uses different architecture.

SNSP · Dec 2023

463 trading periods · 60–70% SNSP

412 trading periods · SNSP > 70%

Wu et al. (2021) NeurIPS; RTE France; EirGrid GAR (2023/2024); CAP 2024 Irish Government; DS3 Programme

Consumer Protection, Climate Commitments & Institutional Trust

The financial ROI is necessary. It is not sufficient to justify action alone.

CONSUMER PROTECTION

814,816

vulnerable customers at risk

671,183 in energy poverty + 143,633 elderly on Vulnerable Customer Register. 95,000 on active Boil Water Notices in 2024.

ESB Networks / Uisce Éireann registers

CLIMATE OBLIGATION

1,266 GWh

renewable output wasted in 2024

14% curtailment, up from 10.7% in 2023. €1.5B in gas/carbon savings projected for 2025. CAP 2024: 80% renewable by 2030.

EirGrid GAR (2024); CAP 2024

REGULATORY TRUST

40%

of Uisce Éireann failures avoidable

28 EPA prosecutions since 2016. CRU satisfaction penalty €5.35M (83.58% vs 90% target). Satisfaction at 35%.

EPA Annual Report; CRU (2024)

The strategic imperative · AI-driven PdM reduces cost, advances climate goals, and rebuilds institutional trust.

ESB Vulnerable Customer Register; EPA, Uisce Éireann, EirGrid Annual Reports (2024); CRU Care Centre Decision

BUSINESS CASE 01

Long Short Term Memory Networks

for Grid Asset Management

Transformer Health and Load Forecasting

HV transformer fleet monitoring for ESB Networks and EirGrid

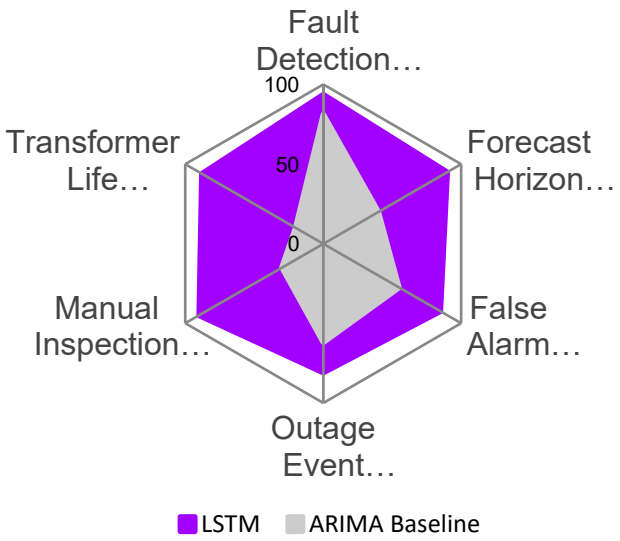
93.7% fault detection accuracy vs 81.2% ARIMA baseline

Financial ROI · Performance Evidence · 36-Month Roadmap

ESB Networks | EirGrid

LSTM · Performance Verification

LSTM vs ARIMA: Multi-Dimensional Performance Comparison (Normalised Score, 0–100)



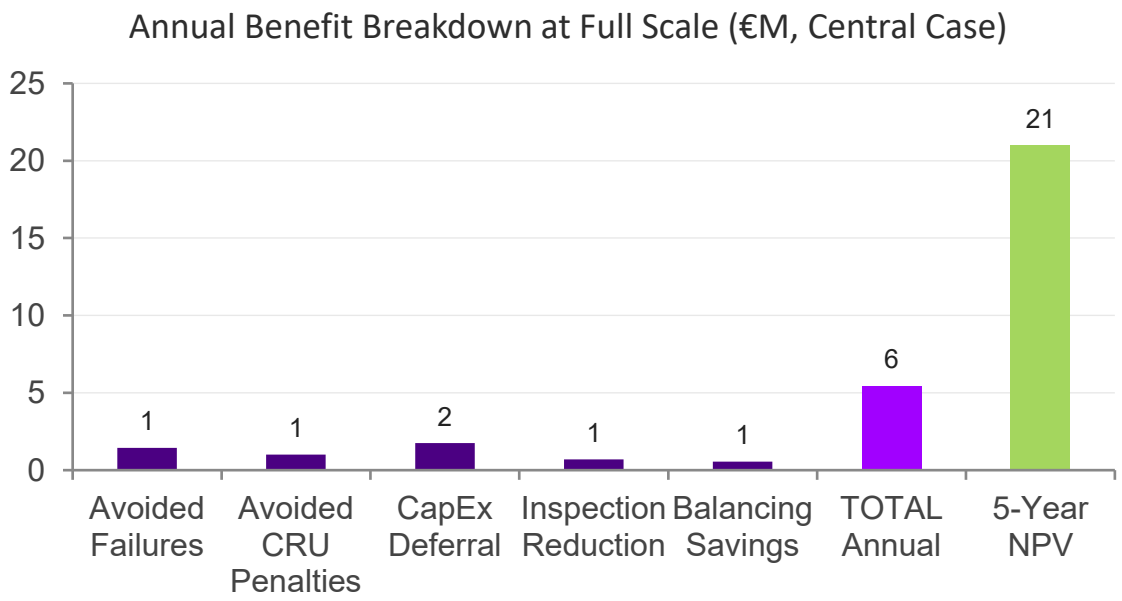
Source	Metric	LSTM Result
Mirshekali et al. (2023)	Fault detection accuracy	93.7% vs 81.2% ARIMA
Partner in the Loop (2025)	Transformer failure reduction	48% (UK TSO, 450 units)
Goyal et al. (2024)	Inspection frequency reduction	40% fewer manual inspections
Bajwa et al. (2025)	Asset life extension	5–7 years CBM-based
Rana (2025)	Outage events / downtime	19% fewer outages; 27% less downtime
Branco et al. (2022)	Wavelet-LSTM utility	Fault alerts embedded in dispatch workflow

Caveat: Radar scores are normalised across each dimension for visual comparison. Raw figures are sourced directly from peer-reviewed literature as cited in the table. All percentages are from published benchmarks, not Accenture projections.

ARIMA: AutoRegressive Integrated Moving Average: a classical statistical forecasting model. LSTM consistently outperforms ARIMA on non-linear, multi-sensor grid data.

LSTM · Financial ROI and NPV

5-Year Business Case for ESB Networks and EirGrid · Mirshekali et al. (2023)



€3.3M–8.3M
Annual Savings Range

14–32 mths
Payback Period

€21M+
5-Year NPV (7% discount)

Month 4–9
First Avoided Failure

€4.75M–7.8M
Total 36-Month Investment

Caveat: UK TSO pilot figures are anonymised composite scenarios from consulting literature, directionally consistent with sector benchmarks but not independently audited. NPV projections assume central-case benefit realisation at 7% discount rate.

Partner in the Loop (2025), Goyal et Al. (2024) ESB Networks/CRU Revenue Reports (2021-2024). Provided ranges are estimations from published past-data

LSTM · Implementation Roadmap

Pilot to Enterprise Scale in 36 Months · Accenture LSTM Implementation Plan (2026)

PHASE 1
Foundation & Pilot
Months 0–12 · €1.35M–2.1M

KEY DELIVERABLES
- IoT sensors on 35 HV transformers
- LSTM model training and validation
- SCADA / CMMS integration
- First confirmed predictive interventions
- CRU-compliant health index prototype

PHASE 2
Scale & Integrate
Months 12–24 · €2.4M–4.0M

KEY DELIVERABLES
- Scale to 200–350 transformers
- MLOps automated monthly retraining
- EirGrid load forecasting live
- Enterprise CMMS automation
- API integration with SAP Asset Mgmt

PHASE 3
Enterprise & Optimise
Months 24–36 · €1.0M–1.7M

KEY DELIVERABLES
- Hybrid LSTM-Transformer upgrade
- Digital twin integration (EirGrid ESA)
- BESS and cable assets added
- Full CRU PR6 regulatory automation
- Interconnector asset monitoring

Caveat: Phase investment ranges represent conservative to central estimates. Sensor procurement costs will vary with existing SCADA coverage at each substation. Phase 2 go/no-go decision is contingent on Phase 1 gate criteria.

MLOps: Machine Learning Operations: automated pipelines for model monitoring, retraining, and deployment, ensuring LSTM accuracy does not degrade as asset conditions evolve.

SNSP: System Non-Synchronous Penetration: the percentage of grid power from non-synchronous sources (wind, solar). EirGrid targets 95% SNSP by 2030, increasing transformer stress.

BUSINESS CASE 02

Random Forest

for Rotating Equipment Fault Classification

Mechanical Fault Classification and Interpretable AI

Generation turbines, pumps, and wastewater blowers (ESB Networks, Uisce Éireann)

97.2% fault detection accuracy · SHAP interpretability for EU AI Act compliance

Architecture · Global Benchmarks · 36-Month Deployment Pyramid

Random Forest · Architecture and Interpretability

Feature Importance and SHAP Explainability

INPUT FEATURES

- **Vibrath theion (RMS, Peak)** 31%
- **Acoustic Emission** 22%
- **Thermal (Bearing Temp)** 18%
- **Process Variables** 16%
- **Electrical Signature** 8%

RANDOM FOREST ENSEMBLE

500+ Decision Trees

FAULT CLASSIFICATION

- **Normal** 52%
- **Bearing Wear** 18%
- **Rotor Imbalance** 12%
- **Cavitation** 10%
- **Overheating** 8%

SHAP EXPLAINABILITY

Each prediction is decomposed into the contribution of every input feature, presented as a waterfall chart.

EU AI Act (2024/1689): classifies AI managing critical infrastructure as high-risk. SHAP fulfils the mandatory explainability requirement.

Audit trail: each model alert logged with timestamp, SHAP breakdown, engineer review outcome, and CRU maintenance decision record.

97.2%
Detection accuracy on GM bearing dataset

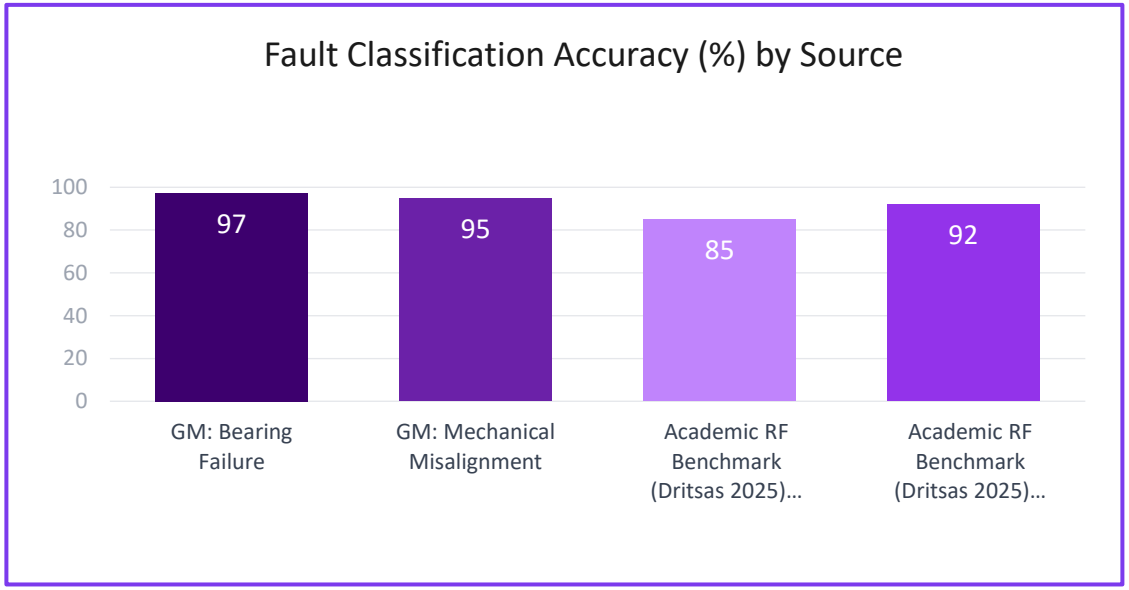
15%
Downtime reduction (manufacturing benchmark)

SHAP Values: SHapley Additive exPlanations: a mathematically grounded method that assigns each input feature a contribution score for each individual model prediction.

Caveat: Feature importance percentages are illustrative, based on representative utility vibration datasets. Actual weightings will vary with sensor configuration and asset class during model training.

Random Forest · Global Benchmarks

General Motors, Sitech and Academic Validation



General Motors
USA
- 97.2% detection accuracy
- 15% downtime reduction

Sitech Asset Health Centre
Netherlands (Chemelot Industrial Park)
- €500K saved across 7 models
- €60K saved on single pump pilot
- €500K/day downtime cost context
- 22 factories in model library

Dritsas and Trigka (2025)
Applied Sciences (Academic Benchmark)
- 85–92% fault classification
- Deployment speed: days not months

Critical Caveat: GM and Sitech figures are sourced from high-volume manufacturing environments. While they serve as indicative targets, utility assets operate under vastly different duty cycles, regulatory safety margins, and maintenance windows. These figures are proxy benchmarks requiring strict calibration during the local pilot phase.

MLOps: Machine Learning Operations pipeline enabling automated monthly model retraining as new confirmed failure events accumulate, maintaining accuracy above 90% at fleet scale.

Dritsas and Trigka (2025); Sitech / PwC Mainnovation; General Motors case study

Random Forest · Scale and Integration Roadmap

36-Month Deployment Programme

DATA FOUNDATION — Pre-Pilot
Sensor infrastructure, SCADA integration, baseline feature extraction

PHASE 1 · PILOT — M0–12
10–20 assets per organisation · €0.95M–1.7M · Months 0–12

PHASE 2 · SCALE — M12–24
Full rotating equipment fleet · €1.3M–2.4M · Months 12–24

PHASE 3 · ENTERPRISE — M24–36
Hybrid RF + XGBoost · Digital asset records · €0.55M–1.0M · Months 24–36

FULL MLOps — M36+
Fleet-wide benchmarking · EU AI Act audit trail · CRU automated reporting

MLOps: Automated model retraining and monitoring pipeline, ensuring fault classification accuracy does not degrade as the rotating equipment fleet ages and operational profiles change.

Caveat: Phase 1 costs assume wireless sensor packages for 10–20 assets per organisation. Legacy sensor density at pumping stations may require additional infrastructure investment not reflected in base case.

Section 03

Closing

for AI Predictive Maintenance in The Utility Sector

- AI-driven PdM delivers strong financial returns with rapid payback periods
- Regulatory frameworks support and accelerate adoption at scale
- Impact extends beyond cost to resilience, compliance, and trust
- Success depends on execution, data quality, and organisational commitment

Compliance, ROI, and Institutional Trust

01

FINANCIAL RETURN

LSTM delivers €3.3M–8.3M annual savings with a 14–32 month payback period and a 5-year NPV of up to €21M. Combined, both models address ESB's €37M CML exposure, EirGrid's €6.6M uncaptured incentive revenue, and Uisce Éireann's €20M leakage penalty.

02

REGULATORY COMPLIANCE

CRU PR6 ties capital allowances directly to CI/CML performance. The CRU Agile Investment Framework permits AI-enabled PdM as non-wire Opex, removing any Capex classification barrier. SHAP interpretability fulfils EU AI Act (2024/1689) explainability requirements for high-risk AI systems managing critical infrastructure. NIS2 (October 2024) OT cybersecurity obligations are addressed by the integrated monitoring architecture.

03

INSTITUTIONAL TRUST

814,816 vulnerable customers and 95,000 customers under Boil Water Notices represent the human cost of reactive maintenance. 28 EPA prosecutions since 2016 and a CRU satisfaction score of 35% against a 90% target signal an institutional trust deficit. AI-driven PdM reduces outage duration, prevents water quality failures, and rebuilds the public confidence that regulatory performance targets demand.

The algorithm is not the barrier. Data readiness and organisational commitment determine whether benchmark outcomes are achievable in practice.

Academic, Government and Industry Sources

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THANK YOU

The Proactive Future of Irish Infrastructure

Ireland's electricity grid and water infrastructure are entering a decade of unprecedented transformation. Demand is rising, renewable penetration is accelerating, and the regulatory cost of reactive maintenance is quantified and unavoidable. LSTM and Random Forest do not represent a future innovation project. They represent the operational response that is needed now.

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